Dials, Not Curtains: Hard-to-Vary Explanations for the Classroom, from a Counterfactual Criterion for Simulations

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Abstract

Schweer and Elstner [1] argue that a simulation explains a phenomenon only when it exposes genuine difference-makers and supports counterfactual reasoning about them, rather than merely reproducing an observed result at whatever level of detail is available. This paper reads that criterion alongside a second, older one: Deutsch's [2] observation that a good explanation is *hard to vary*, in the sense that its details are load-bearing and cannot be swapped for alternatives without destroying its fit to the phenomenon, whereas a bad explanation, however fluent, tolerates arbitrary variation because none of its details were actually doing any work. The synthesis offered here is that these two criteria meet at a single structural feature of a model: an explicit, measurable ratio that separates a regime in which some effect is a genuine difference-maker from a regime in which it is legitimately negligible. Naming such an effect satisfies Schweer and Elstner's requirement; pinning down the precise functional form of its ratio, so that no alternative scaling law would fit the data equally well, is what makes the resulting account hard to vary in Deutsch's sense. Four classroom cases, a spring extending under its own weight, a thermometer equilibrating in a bath, bristles aligning in an electric field suspended in oil, and a sphere settling to its floating depth, are developed to this joint standard, each with an explicit ratio, a predicted curve, and a stated construction by which a class can test that curve against an alternative one that a merely descriptive account could not have ruled out in advance.

1 Introduction

Two literatures converge on the same complaint from different directions. Schweer and Elstner, writing about atomistic simulation in the philosophy of science, worry that a simulation which reproduces an observed outcome in exhaustive microscopic detail may nonetheless fail to explain it, because reproduction alone does not tell you which of the reproduced details actually made the difference to the outcome and which were merely present. Their remedy is a counterfactual test: a simulation explains to the extent that it exposes candidate difference-makers as separately variable quantities, varies them systematically, and shows the outcome tracking that variation. Deutsch, writing about the growth of scientific knowledge quite generally, worries from the opposite side, about explanations that are too easy to hold onto, myths and just-so stories that can absorb almost any modification to their details without changing what they predict, and are for that reason empty even when they are internally consistent and satisfying to tell. His remedy is a rigidity test: a good explanation is one whose details cannot be varied without ruining it, because every detail is functionally tied to the phenomenon it accounts for.

These are not the same test, and a classroom example is more exposed to failing one than the other. A student who says "the thermometer lags because heat has to diffuse into the bulb" has named a difference-maker in Schweer and Elstner's sense: diffusion, as opposed to something else, is doing the explanatory work, and pressed to justify this the student could point to the bulb's material or its immersion in the bath as the relevant variable. But the explanation as stated is still easy to vary in Deutsch's sense, because nothing in it commits to a specific relationship between the bulb's size and the length of the lag; a student could just as easily have said the lag scales with the bulb's surface area, or with its mass, or with the square root of its volume, and the qualitative story would have survived unchanged in each case, which is exactly the symptom Deutsch treats as diagnostic of an explanation that is not yet doing any real work. It is only once the lag is tied to a specific, derivable scaling relationship, in this case $\tau \sim a^2/\alpha$, that alternative guesses about the scaling become distinguishable from the correct one by comparing predicted curves against measured ones, and the explanation becomes hard to vary because almost any other exponent or combination of variables would now produce a visibly wrong curve.

This paper's organising claim is that these two tests are not just compatible but naturally sit on top of one another, and that the site where they meet is the explicit ratio underlying what we will call a regime-dependent idealisation: an idealisation that is not stipulated by convention but genuinely holds in one limit of some measurable parameter and genuinely fails in the other. Section 2 develops this synthesis directly. Sections 3 through 6 work it through the four cases in full, each stated as a specific quantitative prediction, a matching apparatus a class can build or borrow, and a note on precisely how the prediction is hard to vary, meaning what a wrong guess about the underlying relationship would have looked like instead. Section 7 draws the four cases together into a single assessment rubric. An interactive simulation letting a reader turn each of the four ratios directly, and a student-facing task sheet built around the same four cases, accompany this paper as companion materials [4, 5].

2 Two tests, one target

Consider the general shape shared by all four cases below. Some quantity Q, self-weight, internal thermal diffusivity, viscosity, is present in every instance of the phenomenon and never strictly zero, so no honest description can say it is simply absent. There is an outcome of interest, and there is a dimensionless ratio r, built from Q and from the other parameters of the system, such that the outcome depends on Q negligibly when r is in one extreme and depends on it dominantly when r is in the other. A description satisfies Schweer and Elstner's counterfactual criterion the moment it identifies Q and r correctly and states, even qualitatively, that the outcome should change as r is varied across its range: this is already more than a polished description offers, since a polished description would simply report what the outcome was in the one case observed, without committing to what it would have been had r differed.

But qualitative dependence is cheap, and this is where Deutsch's test does further work. To say that "the outcome depends on r" is compatible with a great many different functional forms, linear, quadratic, exponential, logarithmic, and a description that stops at qualitative dependence has not yet ruled any of them out, so it remains easy to vary: swap the true functional form for a wrong one and the description survives unscathed, because the description never committed to a form precise enough to be falsified by the swap. An explanation becomes hard to vary only once it commits to the specific form, $\tau \propto a^2/\alpha$ rather than $\tau \propto a/\alpha$ or $\tau \propto a^2\alpha$, because a specific form generates a specific, checkable curve, and a class equipped with real apparatus can plot the measured curve against the predicted one and see directly whether an alternative exponent would have fit the data at all. The four sections that follow are therefore each built the same way: state the ratio, derive its precise functional form rather than merely asserting a direction of dependence, describe the apparatus, and then say explicitly what a wrong guess at the functional form would have predicted instead, so that the comparison a student is asked to make is a comparison against a genuine, stateable rival, not merely a confirmation of the one curve on offer.

This gives a compact two-axis test for any candidate classroom explanation, which Table 1 states in full and which recurs at the end of every case study below.

	Schweer–Elstner axis	**Deutsch axis**
Question asked	Has a genuine difference-maker been named and separated from the rest of the description?	Is the relationship tying that difference-maker to the outcome specific enough that a wrong guess would visibly fail?
Failure mode	A polished description that reproduces the one case observed without naming what would have changed it.	A qualitatively correct story that would have survived equally well under a different, wrong functional form.
Classroom check	Vary the named quantity and confirm the outcome tracks it.	Predict the specific curve in advance, including a stated wrong alternative, and compare both against data.

Table 1: the joint rubric applied to each case below.

3 The spring: self-weight against applied load

A helical spring of natural length L_0, spring constant k, and total weight W_s distributed uniformly along its length hangs vertically from a fixed support, free at its lower end. The tension at height x measured down from the support must support every coil beneath that point, so for the unloaded spring

$$T_0(x) = W_s\left(1 - \frac{x}{L_0}\right), \qquad 0 \le x \le L_0,$$

a profile that falls linearly from W_s at the support to zero at the free end. Suspend a load W from the free end and every cross-section must now carry the load in addition to the self-weight below it:

$$T(x) = W + W_s\left(1 - \frac{x}{L_0}\right).$$

Since local coil spacing is proportional to local tension, the observable prediction is the ratio of tension at the top to tension at the bottom,

$$\frac{T(0)}{T(L_0)} = \frac{W + W_s}{W} = 1 + \frac{1}{r}, \qquad r \equiv \frac{W}{W_s},$$

which is the quantity a class can actually measure with a ruler against successive coils. Figure 1 plots this predicted tension profile for a self-weight-dominated case ($r = 0.15$) and a load-dominated case ($r = 6$): the two lines have the same slope in absolute tension but represent starkly different fractional variation, nearly a factor of eight top-to-bottom for $r = 0.15$, under fifteen per cent for $r = 6$.

This clears the Schweer–Elstner bar as soon as a student identifies W_s, rather than k or L_0, as the quantity whose neglect is regime-dependent. It clears the Deutsch bar only once the prediction is stated as the specific ratio $1 + 1/r$ rather than a vaguer "heavier loads make the spacing more even," because a wrong but qualitatively similar guess, that fractional variation falls as $1/\sqrt{r}$

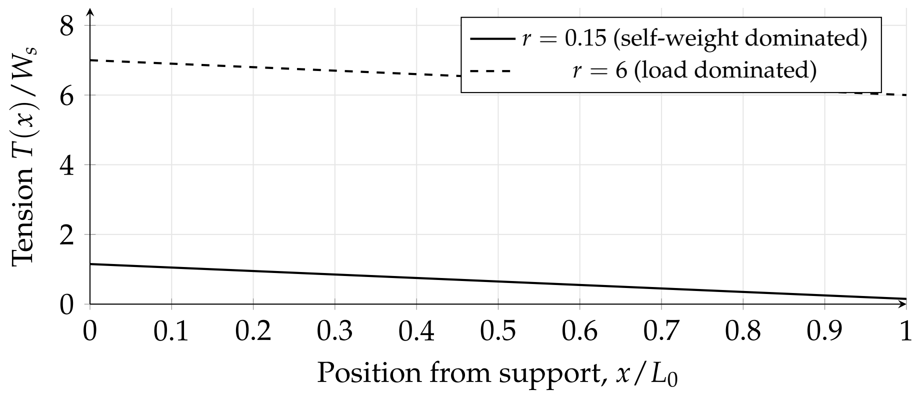

Figure 1: Predicted tension profile $T(x)/W_s = r + (1 - x/L_0)$ along the spring for two values of $r = W/W_s$. The vertical separation of each line across its own length, not its absolute height, is what a class measures as coil-spacing variation.

rather than $1/r$, would also predict "more even at higher load" while giving a visibly different curve, so only a class that plots the measured ratio against r across several loaded trials, rather than checking a single loaded case against a single unloaded one, can distinguish the true linear-in-$1/r$ relationship from that rival.

4 The thermometer: internal diffusion against bath equilibrium

A thermometer bulb of characteristic radius a and thermal diffusivity α, initially at temperature T_i, is plunged into a bath held at T_b. Treating the bulb as a lumped body exchanging heat with an effectively infinite bath gives, to a first approximation used here for its pedagogical transparency rather than its metrological precision,

$$T(t) = T_b - (T_b - T_i)\, e^{-t/\tau}, \qquad \tau \sim \frac{a^2}{\alpha},$$

so the characteristic equilibration time scales with the *square* of the bulb's linear size, not with its size directly and not with its surface area or volume alone. This is the specific, checkable claim: double the bulb's radius and the lag should roughly quadruple, not double.

A commercial thermometer's bulb is too small and its response too fast for this scaling to be watched comfortably by eye, which is precisely why a homemade instrument is the better teaching apparatus here rather than a mere convenience. Take a round-bottomed flask of perhaps 250 ml as the oversized bulb, fit it with a two-holed rubber bung carrying a length of narrow glass or rigid plastic tube as the stem, and fill the flask completely with water dyed with food colouring so that the liquid level sits partway up the tube; the whole assembly is then a crude constant-volume liquid-in-glass thermometer whose "bulb" is roughly two orders of magnitude larger by radius than a laboratory thermometer's, and by the a^2 scaling its equilibration time should be roughly four orders of magnitude longer, comfortably tens of seconds to a few minutes rather than the near-instant response of a shop-bought instrument. Plunge this homemade flask-thermometer into a water bath held a known number of degrees above room temperature, and mark or video the dye column's height at regular intervals; a second, much smaller flask or a small test tube fitted the same way gives the fast-τ comparison case. Figure 2 shows the predicted approach curves for the two bulb sizes on the same bath.

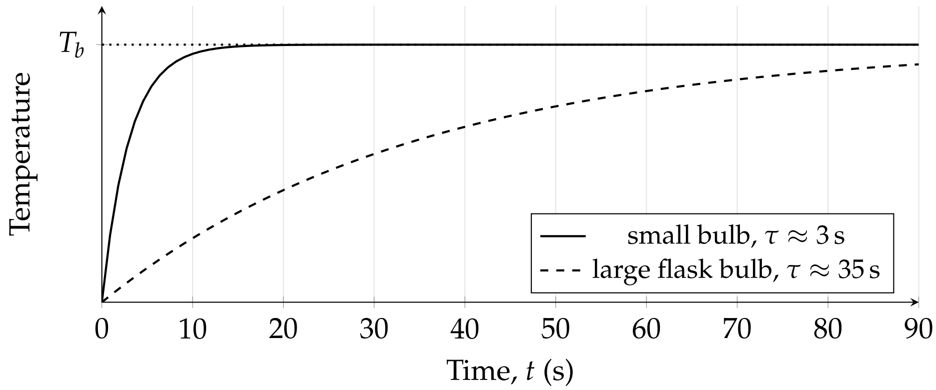

Figure 2: Predicted temperature relative to bath temperature for a small and a deliberately oversized bulb. Both curves approach the same asymptote; only the rate of approach differs, and by the $\tau \sim a^2/\alpha$ scaling that rate should differ by roughly the square of the radius ratio.

The Deutsch test bites hardest here, because the qualitative story, "a bigger bulb takes longer," is nearly unfalsifiable with a single pair of bulbs: almost any monotonic relationship between size and lag would look consistent with one small flask responding faster than one large flask. What makes the account hard to vary is running three or four bulb sizes, plotting measured τ against a on logarithmic axes, and checking that the gradient of that log-log line comes out at 2 rather than at 1 or at $\frac{1}{2}$; a class that has only ever compared two bulbs has satisfied the Schweer–Elstner requirement to name diffusivity as the difference-maker but has not yet earned the stronger, Deutsch-hard-to-vary claim that the specific square-law is correct rather than merely plausible.

5 Fields made visible: bristles in oil between charged plates

Fine bristles or cut hairs suspended in a shallow, viscous, non-conducting oil between electrodes held at high potential difference polarise and rotate to align with the local field direction. The equilibrium orientation depends only on the field, which is fixed by the electrode geometry and voltage; the approach to that orientation is a damped rotation governed by the balance of electrical aligning torque against viscous drag, with angle $\theta(t)$ measured from the initial orientation approximately following

$$\theta(t) = \theta_f - (\theta_f - \theta_0)\, e^{-t/\tau_v} \cos(\omega t) \quad \text{(underdamped, low viscosity)}$$

or a purely exponential approach with no oscillatory term at all once viscous drag is large enough to suppress the cosine term entirely (overdamped, high viscosity), where τ_v and ω both depend on the viscosity, the bristle's moment of inertia, and the field strength. Figure 3 shows a schematic of the two regimes side by side, freshly switched on at left and settled at right, and Figure 4 plots the two predicted angle-against-time curves.

The specific, hard-to-vary claim is not merely that a more viscous oil takes longer, which a student could believe for almost any reason including several wrong ones, but that there exists a critical viscosity below which the approach is genuinely oscillatory, with one or more visible overshoots past the final orientation, and above which no overshoot occurs at all: a class that runs the demonstration in a single oil at a single voltage has satisfied only the qualitative, easy-to-vary version of the story, whereas a class that finds, by trying several oils of graded viscosity, the approximate viscosity at which overshoot first disappears has located a genuine transition

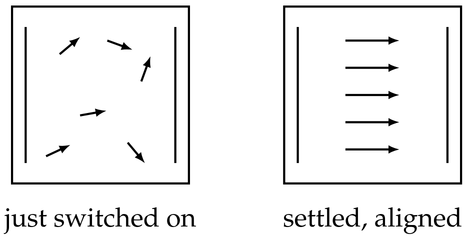

Figure 3: Schematic of bristles between two plane electrodes immediately after switch-on (left, random initial orientations) and once settled (right, aligned along the uniform field). The equilibrium pattern at right is set by the electrode geometry alone; only the time to reach it depends on viscosity.

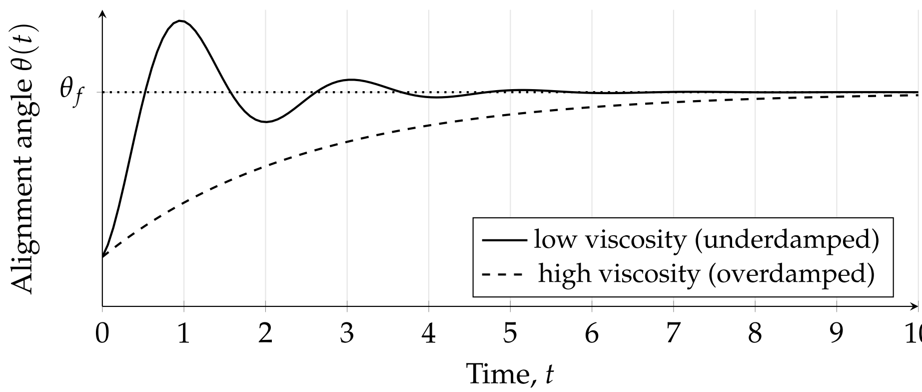

Figure 4: Predicted alignment angle approaching the same final orientation θ_f under low viscosity, with visible overshoot and oscillation, against high viscosity, with a smooth monotonic approach. The final pattern of Figure 3 is identical in both cases.

and can compare it against the predicted critical damping condition rather than against a vague impression of "slower."

6 The floating object: equilibrium depth against approach to it

A sphere of density ρ_s released into a fluid of density ρ_f has an equilibrium floating depth fixed by Archimedes' principle alone, independent of viscosity. Its approach to that depth follows a damped oscillator equation, and depth against time takes the same two functional forms as the bristle's angle against time in Section 5, underdamped with decaying oscillation about the equilibrium depth for a low-viscosity fluid such as water, overdamped with a smooth monotonic approach for a high-viscosity fluid such as glycerol or golden syrup. Figure 5 plots both against the same equilibrium asymptote.

Because Sections 5 and 6 share a functional form, they make a natural paired assessment: a class that has derived the underdamped-versus-overdamped criterion once, for whichever demonstration is run first, should be able to predict, before running the second demonstration, both the qualitative regime it will fall into given the fluids or oils available and the rough number of visible oscillations expected, and a genuinely hard-to-vary account is one where that prediction is stated, and risked, before the tape starts recording rather than fitted to the trace afterward.

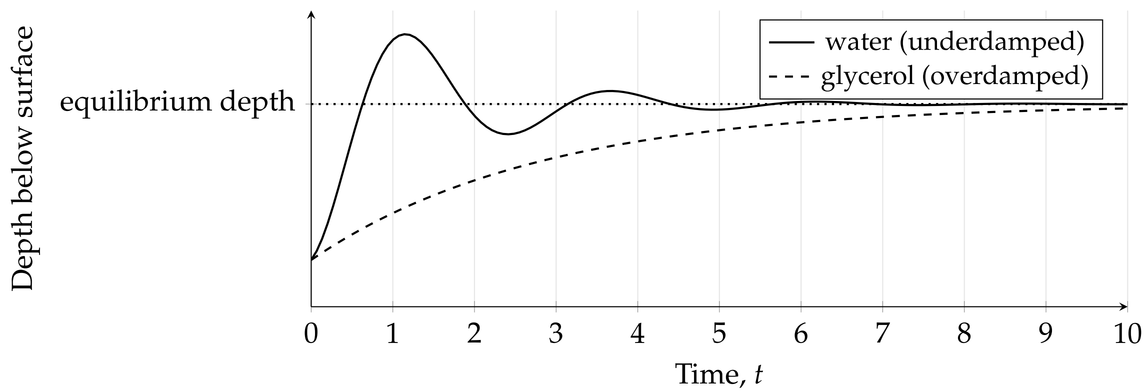

Figure 5: Predicted depth against time for a floating sphere released into a low-viscosity and a high-viscosity fluid. Both traces share the same asymptotic depth, set by Archimedes' principle; only the damping regime differs.

7 A shared rubric

The four cases above are not simply four independent illustrations of a common warning about idealisation; read against Table 1, they are four instances of the same joint test, applied at increasing levels of quantitative commitment. In every case a candidate explanation can be made to satisfy Schweer and Elstner cheaply, by naming the correct quantity, self-weight, diffusivity, viscosity, as a difference-maker and gesturing at a direction of dependence; and in every case that same explanation remains easy to vary in Deutsch's sense until it is pinned down to a specific functional form, a specific exponent, a specific critical ratio, that a class can pit against a stated, plausible-sounding rival and adjudicate with real apparatus. The pedagogical claim this paper is that these two demands are more than "be more careful": the first asks whether the right variable is even in the room, and the second asks whether the relationship claimed for it is precise enough to have been wrong. A classroom explanation that clears only the first hurdle is a fluent description of a single observed case; one that clears both is, in the sense both authors intend, an actual explanation, and it is worth teaching students to notice the difference between the two by asking, of any account they are given or any account they offer, what specific alternative it has ruled out.

References

[1] Schweer, J. and Elstner, M. (2025). *On the explanatory power of atomistic simulations*. European Journal for Philosophy of Science, 15(3), 1–23. `https://doi.org/10.1007/s13194-025-00667-z`

[2] Deutsch, D. (2011). *The Beginning of Infinity: Explanations that Transform the World*. Allen Lane, London.

[3] Woodward, J. (2003). *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press, Oxford.

[4] McCulloch-James, L. (2026). *Dials, Not Curtains: Interactive Companion* [interactive simulation]. McMurmerings. `https://leelemacjames.github.io/dials_of_explanation.html`

[5] McCulloch-James, L. (2026). *Four Dials: Classroom Tasks on Explanation vs. Description* [student task sheet]. McMurmerings. `https://leelemacjames.github.io/dialsOfexplanation.pdf`